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Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Determine whether the series $\sum_{n=0}^{\infty} \sin \frac{(2n+1)\pi}{2}$ converges or diverges. If it converges, find its sum.

Since $\sin \frac{\pi}{2} = 1$ and $\sin \frac{3\pi}{2} = -1$ and the angle may be incremented by any multiple of 2π , then $\sin \frac{(4r+1)\pi}{2} = 1$ and $\sin \frac{(4r+3)\pi}{2} = -1$, (any r).

Examine the partial sums of the series. The expanded series is $\sin \frac{\pi}{2} + \sin \frac{3\pi}{2} + \sin \frac{5\pi}{2} + \sin \frac{7\pi}{2} + \sin \frac{9\pi}{2} + \dots$.

Notice that the angles in consecutive even terms (n is even) differ by 2π , and the angles in consecutive odd terms (n is odd) differ by 2π .

Since $\sin \theta = \sin (\theta + 2m\pi)$, all the even terms are equal to 1, and all the odd terms are equal to -1 .

Evaluating each term, the series is $1 + (-1) + 1 + (-1) + 1 + \dots$. The partial sums alternate between 1 and 0, tending to neither

value. Therefore, $\lim_{k \rightarrow \infty} \sum_{n=0}^k \sin \frac{(2n+1)\pi}{2}$ does not exist and the series diverges.